Appendix

A. Prompt and Parsing Details

This appendix summarizes the prompts and parsing rules used for the direct VLM localization baselines. All direct VLM baselines are evaluated under a unified parsing and post-processing protocol. The purpose of these experiments is to diagnose whether large VLMs can directly perform dense building landmark localization in UAV aerial images without task-specific fine-tuning.

1) **Strict bbox prompt for Qwen3-VL**
For the strict bbox setting, the model is required to directly output building bounding boxes in JSON format. A representative prompt is: "Identify all buildings in this aerial image. Return only a valid JSON object with the key "boxes". Each box must be in the format [x1, y1, x2, y2] in pixel coordinates, where the origin is the top-left corner of the image. The boxes should tightly cover individual building roofs. Do not include roads, vegetation, shadows, background regions, or partial inaccurate boxes. If there is no building, return {"boxes": []}. Do not output explanations, markdown, or any text outside the JSON." The raw strict prompt often fails to enumerate dense building instances and sometimes returns malformed or incomplete JSON. Therefore, we further evaluate an enhanced strict prompt-v2.

2) **Enhanced strict bbox prompt-v2**
The enhanced prompt-v2 strengthens three constraints: output format, dense enumeration, and box tightness. A representative version is: "Detect every visible building roof in the aerial image. Output only a valid JSON object:
{
"boxes": [
[x1, y1, x2, y2],
...
]
}
Coordinates must be pixel coordinates in the image, with x increasing to the right and y increasing downward. Each box must tightly cover one building roof. Do not merge multiple buildings into one box. Do not output roads, trees, cars, shadows, background, or incomplete building fragments. If no building is visible, output { "boxes": [] }. The answer must be valid JSON only, without markdown or explanations." This prompt improves the parse success rate and the localization performance of Qwen3-VL compared with the raw strict prompt, but the resulting COCO AP remains far below proposal-based and supervised detection methods.

3) **Flexible point/box prompt**
For flexible point/box evaluation, the model is allowed to output either building center points or bounding boxes. This setting is designed to test whether VLMs can at least recognize approximate building centers even when precise bounding boxes are difficult. A representative prompt is: "Find all buildings in this aerial image. For each building, output either its center point or its bounding box. Return only a valid JSON object with the key "objects". Each object may contain:
{
"point": [x, y],
"bbox": [x1, y1, x2, y2]
}
Coordinates must be pixel coordinates. If only a center point is available, provide the point. If a tight box is available, provide the bbox. Do not output non-building regions. Do not include any text outside the JSON." For flexible outputs, the main metric is Point-AP. Pseudo COCO AP is reported only as a diagnostic after converting point-like predictions into pseudo boxes; it is not directly comparable to standard bbox AP.

4) **Molmo English prompt-v2**
For Molmo, we use an English flexible prompt because it follows English localization instructions more reliably in our experiments. A representative prompt is: "Look at the aerial image and locate all building roofs. Return a JSON list of building locations. Each building should be represented by either a center point [x, y] or a bounding box [x1, y1, x2, y2] in image pixel coordinates. Output JSON only. Do not include explanations. Do not mark roads, vegetation, shadows, background, or non-building objects." The Molmo flexible English prompt-v2 improves Point-AP compared with the raw Molmo flexible prompt, but its point-level recognition remains much weaker than proposal-based methods.

5) **JSON repair and parsing rules**
Since direct VLM outputs are often not strictly valid JSON, we apply a unified JSON repair and parsing protocol before evaluation. First, markdown fences, code block markers, leading explanations, and trailing comments are removed. If the model outputs multiple text segments, the parser extracts the first bracket-matched JSON object or JSON array. Common key names are normalized, including `boxes`, `bbox`, `bboxes`, `detections`, `objects`, `points`, and `centers`. Numeric strings are converted to floating-point values when possible. Predictions with missing coordinates, non-numeric values, or invalid length are discarded. For bbox outputs, both [x1, y1, x2, y2] and [x, y, w, h] style boxes are handled when the format can be inferred. Coordinates are clipped to image boundaries. Boxes with non-positive width or height are discarded. When confidence scores are not provided by the VLM, predictions are ranked according to the output order with monotonically decreasing scores.

6) **Coordinate normalization rules**
The parser supports both absolute pixel coordinates and normalized coordinates. If all coordinate values are in [0, 1], they are interpreted as normalized coordinates and multiplied by the image width or height. If coordinates appear to be percentages, they are converted to pixel coordinates accordingly. All final coordinates are clipped to the valid image range. For point outputs, a point is considered valid if it lies inside the image. Point predictions are evaluated using the unified point

recognition protocol. For pseudo bbox diagnostics, point-only predictions are converted to small pseudo boxes by the evaluation script; these pseudo AP values are used only to indicate rough localization quality and should not be interpreted as standard bbox AP.

7) **Duplicate removal and NMS rules**
For direct bbox outputs, duplicate and highly overlapping predictions are removed using non-maximum suppression (NMS) before evaluation. For proposal-based methods, duplicate suppression follows the locked post-processing configuration described in Section III-H of the main text. Specifically, RV-VLM-LA-BR uses the score fusion and Gaussian Soft-NMS defined in the main method section. The locked inference parameters are α =0.4, Soft-NMS IoU threshold 0.1, σ =2.0, and at most 100 detections per image. This appendix only describes the parsing-side duplicate handling for direct VLM outputs.

B. Post-processing and Cross-domain Diagnostic Details

The main progressive component analysis is reported in Table V of the main paper. This appendix provides additional post-processing validation and cross-domain diagnostic details that are not fully expanded in the main text.

1) **Soft-NMS post-processing validation**
To verify the locked post-processing pipeline, we compare two Gaussian Soft-NMS sigma values on the WHU first-50 subset using the same raw proposal scores. The final evaluation uses score fusion α=0.4, Soft-NMS IoU threshold 0.1, maxDets = 100, and σ=2.0, which best reproduces the locked first-50 metrics. The σ=2.0 setting closely matches the locked first-50 reference in AP50:95, AP50, and AR@100. Therefore, the final post-processing configuration uses Gaussian Soft-NMS with σ=2.0.

TABLE B.1
SOFT-NMS SIGMA VALIDATION ON WHU FIRST50

Post-processing	AP50:95	AP50	AP75	AR@100
Gaussian Soft-NMS, sigma=2.0	0.5524	0.7541	0.631	0.7021
Gaussian Soft-NMS, sigma=20.0	0.5474	0.7351	0.6326	0.704
Locked first50 reference	0.5523	0.7558	0.622	0.7024

2) **Inria BoxRefine diagnostic discussion**
The full Inria cross-dataset diagnostic results are reported in Table IV of the main paper. The key diagnostic variant is RV-VLM-LA-BR w/o BoxRefine at inference, which keeps the same WHU-trained region verifier scores as RV-VLM-LA-BR but uses the original UPN proposal boxes instead of refined boxes. This diagnostic variant outperforms the refined-box version on Inria. This suggests that the WHU-trained region verifier transfers well across datasets, while the WHU-trained BoxRefinementHead is more sensitive to cross-domain geometric shifts. Therefore, the Inria results indicate that proposal re-ranking and region verification provide the main cross-dataset gain, whereas multi-domain or domain-adaptive box refinement remains an important direction for future work.

3) **Locked inference hyperparameters**

TABLE B.2
LOCKED INFERENCE HYPERPARAMETERS

Hyperparameter	Value
Max UPN proposals per image	128
Score fusion exponent alpha	0.4
Inference refine lambda	0.75
Soft-NMS type	Gaussian
Soft-NMS IoU threshold	0.1
Soft-NMS sigma	2
Max detections per image	100
Min box size	2 pixels
Inria hyperparameter tuning	No

Table B.2 summarizes the locked inference configuration used in the final WHU and Inria evaluations. These settings are selected on the WHU validation split and are not tuned on Inria. We do not tune α, refine λ, or Soft-NMS parameters on the Inria evaluation set. This preserves the no-Inria-fine-tuning setting and makes the Inria experiment a cross-dataset diagnostic rather than a test-time hyperparameter search.